Unit – 3
Mixed Reality
Tangible Interactions in MR

End use-cases best suited for MR

- Tangible Interactions in Mixed Reality (MR) refer to a method where users interact with digital (virtual) objects using physical objects or real-world tools.

- Instead of only using a controller, keyboard, or touchscreen, users can touch, move, or manipulate real objects to control virtual elements in a Mixed Reality environment.

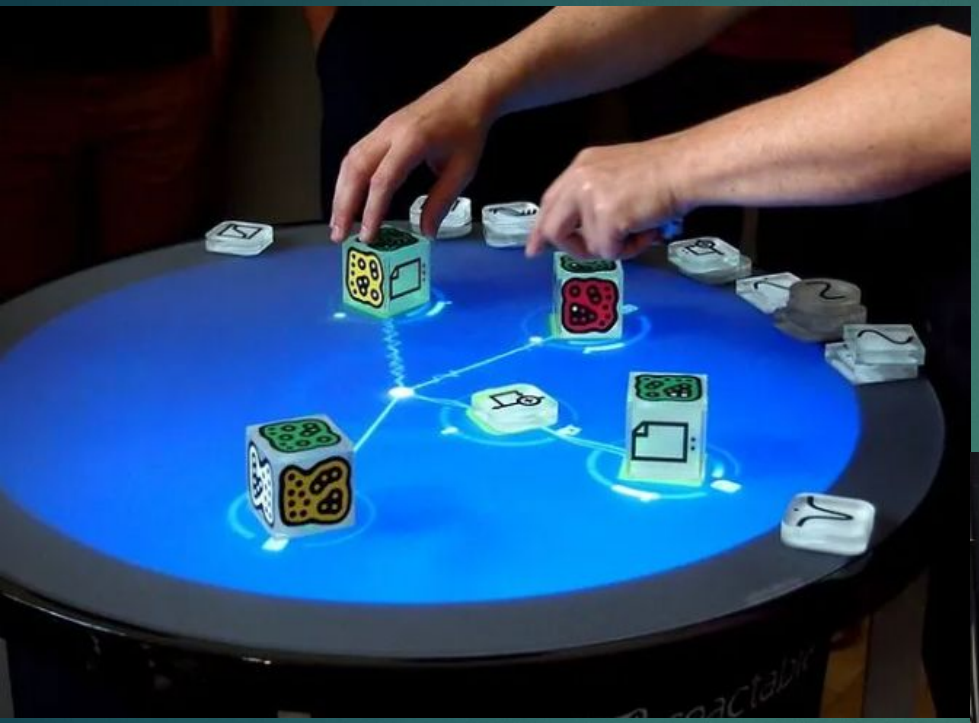
Definition

- Tangible Interaction in Mixed Reality is an interaction technique in which physical objects are used as interfaces to manipulate digital information in a mixed reality environment.
- In simple words:
  In Real world, control digital objects.
- Example:
  A physical cube placed on a table can represent a 3D model in MR, and rotating the cube rotates the virtual model.

Key Characteristics of Tangible Interactions

- Physical Manipulation
  - Users handle real-world objects such as blocks, tools, or models.
- Direct Interaction
  - Interaction feels natural because a person can handle physical objects and monitor the virtual presentation.
- Real-Time Feedback
  - The virtual environment responds immediately to physical actions.
- Spatial Interaction
  - The position and movement of objects in real space affect digital content.
- Intuitive Learning
  - Users require minimal training because interaction is based on real-world behaviour.

How Tangible Interaction Works in MR



Explanation:

- **Physical Object**
  - A real object such as a marker, tool, or controller.
- **Sensors / Camera**
  - Devices capture the movement or position of the object.
- **Tracking System**
  - Software detects and tracks the object.
- **MR Software**
  - Interprets the object's movement.
- **Virtual Response**
  - The digital object changes accordingly in the mixed reality environment.

Simple Example Scenario

- Imagine wearing a Mixed Reality headset such as **Microsoft HoloLens**.
  - You place a **physical cube on a table**.
  - The system recognizes the cube.
  - A **virtual 3D building** appears on top of it.
  - When you rotate the cube, the **building rotates in MR**.
- This is **Tangible Interaction** because **the real cube controls the virtual object**.

Components of Tangible Interaction in MR

- **Physical Objects**
  - These are the **tangible elements** used by users.
  - Examples:
    - Physical blocks
    - Tools
    - Cards
    - Controllers
    - Models

Components of Tangible Interaction in MR

- **Sensors and Cameras**
  - Sensors detect the **position**, **movement**, **and orientation** of the physical objects.
  - Common technologies:
    - Cameras
    - Motion sensors
    - Depth sensors
    - RFID tags
    - Markers

Components of Tangible Interaction in MR

- **Tracking System**
  - The tracking system identifies the **location and movement of objects in real time**.
  - Tracking methods include:
    - Marker-based tracking
    - Marker-less tracking
    - Sensor-based tracking

Components of Tangible Interaction in MR

- **MR Software**
  - Software processes the input from sensors and translates it into **virtual actions.**
  - Functions include:
    - Object recognition
    - Interaction mapping
    - Rendering digital content

Display System

- Displays show the **mixed environment** where digital objects appear integrated with the real world.

- Example devices include **Microsoft HoloLens** and **Magic Leap 1**.

Thank You…!!!